Zifeng Zhang

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Academic Appointments

Postdoctoral Associate 2025–Present
Department of Biostatistics, Yale School of Public Health, New Haven, CT
Postdoctoral Mentor: Prof. Hongyu Zhao

Education

Ph.D. in Statistics, Colorado State University 2025
Dissertation: *New developments on linear regression with random design and high-dimensional mediation analysis*
Advisor: Prof. Wen Zhou; Co-advisor: Prof. Haonan Wang

M.A.S. in Applied Statistics, Nankai University, Tianjin, China 2018

B.A. in Economics, Specialization in Insurance, Nankai University, Tianjin, China 2016

Research Interests

- High-dimensional statistics
- Robust statistical inference under complex dependence
- Causal inference and high-dimensional mediation analysis
- Statistical learning with embeddings and biomedical applications

Publications and Manuscripts

1. **Zhang, Z.**, Ding, P., Zhou, W., and Wang, H. (2025). "With Random Regressors, Least Squares Inference Is Robust to Correlated Errors with Unknown Correlation Structure." *Biometrika*, 112(1), asae054. [Journal] [arXiv]

2. **Zhang, Z.**, Ding, P., Zhou, W., and Shi, C. (2026). "A Duality of the F Distribution." [arXiv]

3. **Zhang, Z.**, Lei, L., Ding, P., and Zhou, W. "Two Routes to Validity: Why Nonstandard Standard Errors Are More Robust Than We Think." Manuscript available upon request.

4. **Zhang, Z.**, Yang, F., Ding, P., and Zhou, W. "High-Dimensional Mediation with Treatment–Mediator Interactions: Signal-Induced Ill-Conditioning and Inference." Manuscript available upon request.

5. **Zhang, Z.**, Wang, L., Huo, Z., Wu, P., Liu, Y., Liu, D., and Zhao, H. "Enhancing Prognostic Accuracy in Idiopathic Pulmonary Fibrosis: A Novel Momentum Feature for Longitudinal Biomarker Analysis." Manuscript available upon request.

Research in Progress

6. **Zhang, Z.**, Tian, Y., Feng, Y., and Zhao, H. "Embedding-Assisted Elastic Net."

7. Magid-Bernstein, J., **Zhang, Z.**, Grychowski, L., Hadiwidjaja, B., Sorensen, G., Remillard, W., Lidman, A., Vargas, D., Falcone, G. J., Sheth, K. N., Gilmore, E. J., Matouk, C., Petersen, N., Lyman, K., O'Keefe, L., Kim, J., Zhao, H., and Sansing, L. H. "Distinct Cytokine Trends Are Associated with Outcome after Aneurysmal Subarachnoid Hemorrhage."

Teaching Experience

Independent Instructor, Colorado State University

STAT 301/307: Introduction to Applied Statistics Fall 2021–Spring 2025 (8 semesters)
Responsibilities: Independently taught the semester-long course, delivered lectures, assessed student learning, and provided office hours.
Course content: Descriptive statistics, statistical inference, power, ANOVA, and simple and multiple linear regression.
Class size: Approximately 50–60 students per class.

STAT 201: General Statistics Summer 2020, Summer 2024
Responsibilities: Independently taught the summer course, delivered lectures, assessed student learning, and supported students outside class.
Course content: Descriptive statistics, statistical inference, and simple linear regression.
Class size: Approximately 10 students per class.

Recitation Instructor, Colorado State University

STAT 201: General Statistics Fall 2019, Spring 2020, Spring 2021
Responsibilities: Led problem-solving sessions, reviewed lecture material, and guided students through exercises and questions.
Course content: Descriptive statistics, confidence intervals, hypothesis testing, and simple linear regression.
Class size: Three recitation sections, with approximately 25 students per section.

Teaching Assistant, Colorado State University

Responsibilities: Graded homework and selected examination problems in undergraduate and graduate statistics courses; also helped prepare homework solutions for STAT 720.

STAT 730: Advanced Theory of Statistics Spring 2023, Spring 2024, Spring 2025
Course content: Advanced statistical theory and inference.

STAT 720: Probability Theory Fall 2022, Fall 2023, Fall 2024
Course content: Graduate probability theory, measure theory, and convergence.

STAT 520: Introduction to Probability Theory Fall 2021
Course content: Probability, random variables, distributions, and expectation.

STAA 573: Analysis of Time Series Fall 2020
Course content: Time-series methods, dependence, and forecasting.

STAA 551: Regression Models and Applications Fall 2020
Course content: Regression modeling, inference, and model assessment.

STAT 315: Intro to Theory and Practice of Statistics Fall 2019
Course content: Foundations of statistical theory and applied statistical methods.

Invited Talks

- "Dual Validity in Linear Regression: Rethinking Robust Inference When Error Structures Are Unknown." ASA Section on Statistical Learning and Data Science Conference (SLDS), New York University, New York, NY, November 2026. *Invited talk, forthcoming.*
- "Dual Validity in Linear Regression: Rethinking Robust Inference When Error Structures Are Unknown." New England Statistics Symposium (NESS), University of Connecticut, Storrs, CT, May 2026. *Invited talk.*
- "Dual Validity in Linear Regression: Rethinking Robust Inference When Error Structures Are Unknown." New Jersey Institute of Technology, Newark, NJ, December 2025. *Statistics seminar.*

Contributed Presentations

- "When Least Squares Inference with Random Regressors Encounters Unknown Correlated Errors." Western North American Region Conference (WNAR), Fort Collins, CO, June 2024. *Student paper oral presentation.*
- "Mediation Analysis with Interactions and Heredity under Growing Number of Mediators." Institute for Research in Statistics and its Applications (IRSA), Minneapolis, MN, May 2024. *Student Poster Competition.*

Academic Service

Journal Referee
Biometrika

Journal Club Organizer
Zhao Lab Journal Club, Yale School of Public Health, Summer 2026
Coordinated weekly meetings and research discussions; presented papers on foundation models and statistical watermarking for large language models.

Statistical Consulting and Collaborative Data Analysis
YIV-906 (CALM Trial) biomarker analysis, 2026
Contributed statistical analyses of clinical laboratory biomarkers, treatment-group comparisons, and time-to-event outcomes in an interdisciplinary study.

Honors and Awards

Graduate Scholarship for Outstanding Freshmen — Nankai University, China — 2016
Gongneng Scholarship — Nankai University, China — 2013

Professional Memberships

New England Statistical Society (NESS), Regular Member — 2026–Present

Computing Skills

R; LaTeX; Slurm; high-performance and parallel computing